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Smart Cities

“A smart city is an urban area where the technology and data collection help improve quality of life as well as the sustainability and efficiency of city operations” (Gomstyn & Jonker, 2025).

Smart cities use IoT (Internet of Things) devices like smart meters, lights, and sensors to collect and analyze data. This data is used to improve everyday life and activities, increase public safety, or improve the environment. Some of the areas where smart technologies play an important role include transportation, infrastructure, and energy.

Smart cities are also positioned as a key strategy for sustainability, since they aim to make urban areas more sustainable by maximizing recycling and reuse, using sustainable resources, extending the useful life of products, or promoting sharing platforms.

Smart Traffic Management

Traffic management refers to the organization and guidance of a traffic system that includes moving and stationary traffic as well as traffic lights and signals, pedestrians, and cyclists. The goal of AI in traffic management is to establish a traffic network that functions like clockwork. It’s in this field is very recent and based on the collection and analysis of real-time data.

Some use cases of AI in traffic management are:

- Traffic flow prediction: By understanding traffic analyzing real-time and historical data, AI can predict congestions and help cities act before delays build up.
- Incident detection and management: AI can spot unusual road events within seconds, allowing traffic operators to respond immediately.
- Adaptive traffic signal control: AI-powered signals that will adjust in real time based on the needs of the traffic flow, ensuring smoother traffic flow.

Urban Energy Management

The demand for efficient and sustainable energy is increasing, and AI is enabling cities to optimize energy consumption and integrate renewable resources. One of the most important applications of AI in this field is the AI-Driven Smart Grids that are revolutionizing urban energy management. These grids utilize machine learning algorithms to predict energy demand and optimize energy flow. Los Angeles has reduced energy losses by 20%.

Privacy and Data Use

With the integration of AI in cities, for example, into traffic management systems. A lot of data privacy concerns arise. Smart cities collect vast amounts of data like real-time traffic information, vehicle details, and personal data through cameras and IoT devices. This constant data collection creates risks around data privacy. Safeguarding this data from unauthorized access and ensuring that data privacy standards are the main objectives. One key safeguard is data anonymization, which removes or masks personally identifiable information so that individuals cannot be easily traced back to the data collected about them.

Bias and discrimination

Another significant ethical concern in AI-driven smart cities is racial bias in facial recognition systems used for surveillance and law enforcement. Where the main affected are people of color, the elderly, and individuals overrepresented in policing databases. As a result, biased facial recognition has led to real-world consequences like false arrests, biased policing, and systemic discrimination. To address this ethical issue, Smart cities and technology developers must prioritize training AI models on more diverse and representative datasets and enforce stronger regulations.

Future Trends

In the next 5 to 10 years Smart Cities are going to shift from isolated systems to a fully integrated AI network. For example, instead of traffic security and public services functioning isolated from each other, AI will build workflows that connect dynamically to all these domains. Another great advance is Digital twins, virtual models of city infrastructure that will also become more common, enabling planners to simulate the impact of new policies or infrastructure projects before implementing them in the real world. This interconnected system will also introduce new risks that Smart Cities will need to prepare for. For example, cybersecurity threats will become a greater concern, since vulnerabilities will affect multiple critical services at once. There are also risks of inequality, due to some neighborhoods, schools, or organizations that cannot afford these systems.

In conclusion, smart cities will likely become the new normal everywhere in the world. Ten years from now, it will probably be hard to imagine a city without AI improving traffic through real-time analysis. But I believe there are many ethical and cybersecurity concerns that we need to be careful about, since they could lead to serious problems.

For example, I worry about traffic management and the mistakes the system could make. Imagine a situation where an AI error causes two cars to crash. On the cybersecurity side, even though there are already cyberattacks from terrorists and military groups today, these attacks would no longer stay in the world of the internet or government systems, they would have real, direct effects on people's everyday lives.

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